KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN

(Permanently Affliated to Thiruvalluvar University)

(Accredited by NAAC & An ISO 9001:2015 Certified Institution)

Thenmathur,Tiruvannamalai

The Project is Submitted to the Thiruvalluvar University, Vellore in partial fulfillment of the requirements for the award of project

Degree of

BACHELOR OF SCIENCE IN ARTIFICIAL INTELLIGENCE

BY

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On the topic of

STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM(SECMS)

STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM (SECMS)

User Story:

The Student Enrollment & Course Management System is a comprehensive Salesforce application designed to streamline and automate the academic enrollment operations of an educational institute. This system enables administrators, enrollment officers, and instructors to efficiently manage student records, course offerings, enrollments, instructor assignments, approval processes, payments, and reporting — all within a unified Salesforce environment.

1. INTRODUCTION

1.1 Project Overview:

This project aims to centralize all student, course, and enrollment-related data within a single Salesforce application to ensure easy data access, structured workflows, and accurate reporting. The system allows Enrollment Officers to manage student admissions, course registrations, and communication effectively, while instructors and administrators gain full transparency into course allocations and academic progress.

The solution enables automated instructor assignment, task creation, validation checks, and multi-level approval processes for enrollment management. With Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the organization can ensure that all enrollment operations — from admission to course completion — are handled systematically with continuous monitoring of key performance indicators such as enrollment volume, revenue, student status distribution, and pending approvals.

1.2 Purpose:

The primary purpose of SECMS is to replace manual and fragmented enrollment processes with a scalable, automated, and secure Salesforce solution that ensures:

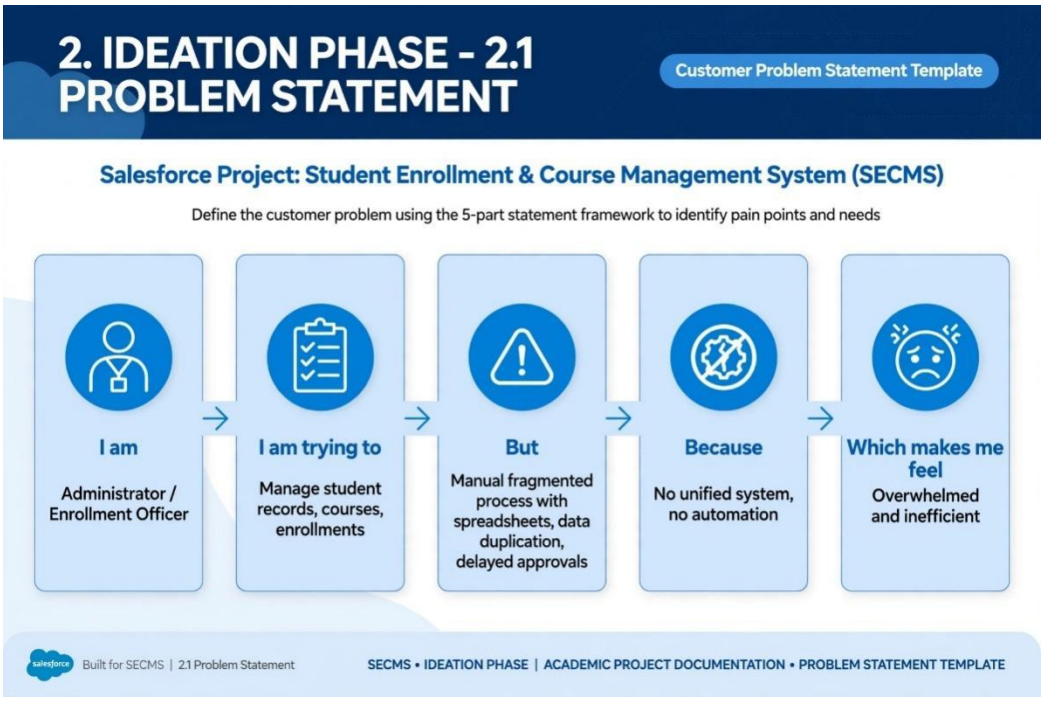
- Accurate student data management

- Structured enrollment workflows

- Automated approvals and validations

- Transparent reporting and monitoring

2. IDEATION PHASE



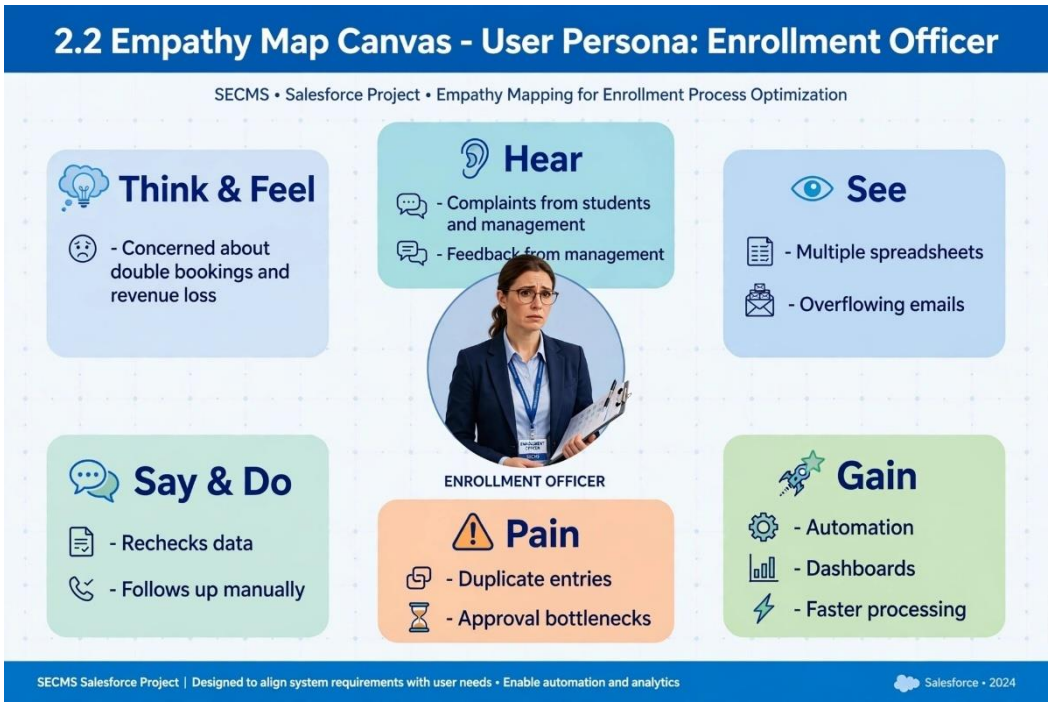
2.1 Problem Statement (Customer Problem Statement Template):

PS No I am (Customer) I'm trying to But Because Which makes me feel

PS-1 Enrollment Officer Manage student enrollments efficiently Enrollment data is scattered across systems Systems are manual and unintegrated Frustrated and error-prone

PS-2 Administrator Track enrollments and revenue Reporting is delayed and inaccurate Data is not centralized Uncertain in decision-making

2.2 Empathy Map Canvas - User Persona: Enrollment Officer



Role: Enrollment Officer with 3-5 years of experience in University Admissions.

- Think & Feel: Concerned about double bookings, data inaccuracies, and potential revenue loss. Seeks reliability and automation.

- Hear: Complaints from students regarding delays and pressure from management to close admissions quickly.

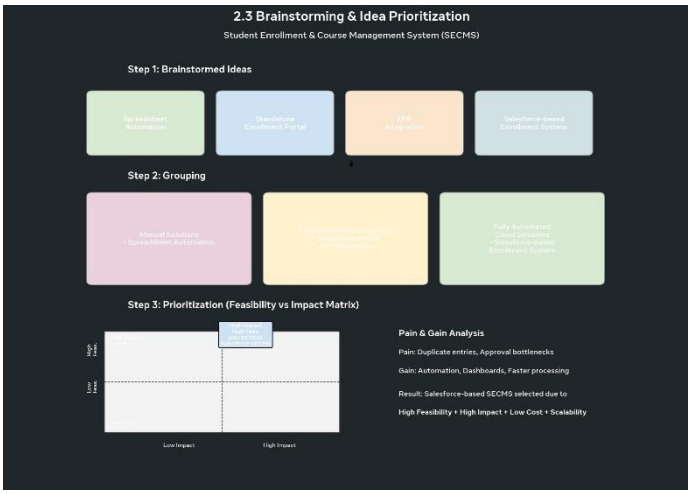
- See: Multiple spreadsheets, emails, and inconsistent data with no single source of truth.

- Say & Do: Rechecks data manually to avoid errors, follows up on approvals via email and calls.

- Pain: Duplicate entries causing confusion and approval bottlenecks delaying enrollment.

- Gain: Automation to eliminate manual work, real-time dashboards for visibility, and faster processing.

2.3 Brainstorming & Idea Prioritization:



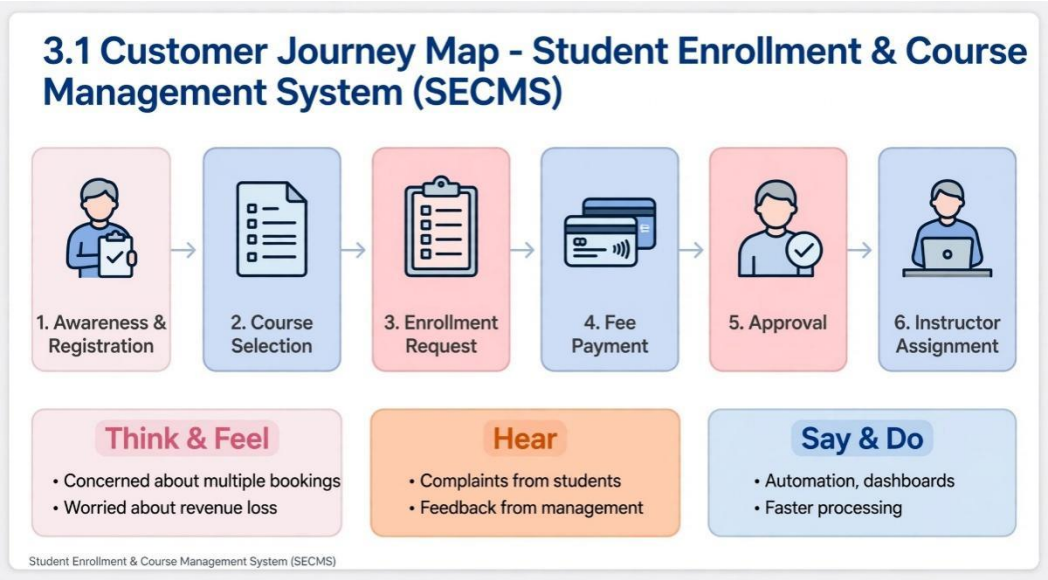
- Step 1 - Brainstormed Ideas:Spreadsheet automation, Standalone enrollment portal, ERP integration, Salesforce-based enrollment system.

- Step 2 - Grouping: Manual solutions, Semi-automated solutions, Fully automated cloud solutions.

- Step 3 - Prioritization: Salesforce-based SECMS was selected due to High Feasibility and High Impact on the feasibility-impact matrix.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map:



The end-to-end journey covers: Awareness & Registration -> Course Selection -> Enrollment Request -> Fee Payment -> Approval -> Instructor Assignment -> Reporting & Completion.

3.2 Solution Requirements:

Functional Requirements:

- FR-1 Student Management: Manage student records (personal, contact, status)

- FR-2 Course Management: Create and manage courses, fees, categories

- FR-3 Enrollment Processing: Request, approve, reject enrollments

- FR-4 Instructor Assignment: Auto-assign instructors based on course category

- FR-5 Reporting: Real-time reports and dashboards for monitoring

Non-Functional Requirements:

- NFR-1 Usability: Simple, intuitive Lightning interface with clearly labeled tabs and role-based dashboards. Ex: Quick registration form, one-click approval.

- NFR-2 Security: Confidentiality and protection via Profiles, Roles, and OWD. Ex: Only admins modify courses, instructors view only assigned students.

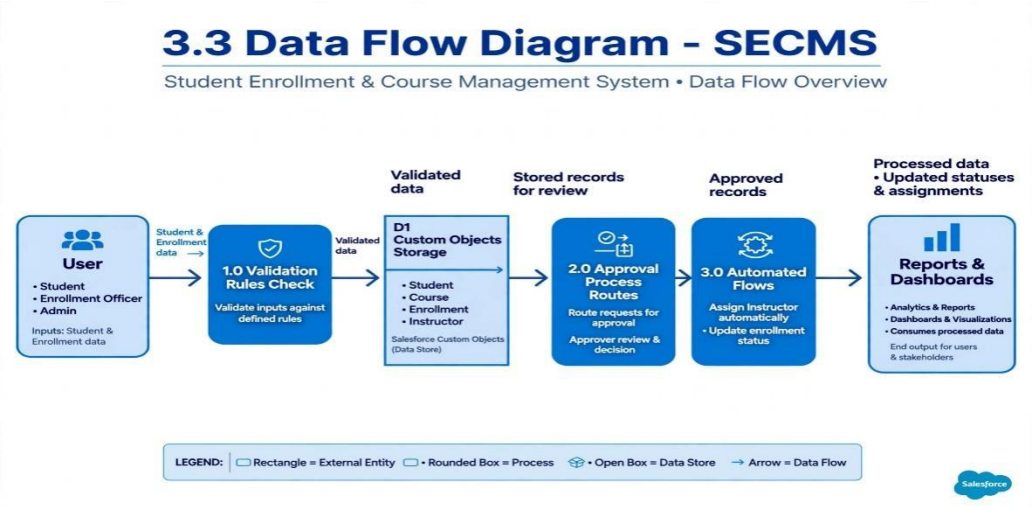
- NFR-3 Reliability: Consistent operation without failures. Ex: No duplicate registration, accurate status updates.

- NFR-4 Performance: Fast response times during peak admission. Ex: Quick dashboard refresh, fast student search.

- NFR-5 Scalability:Support growth in students, courses, departments without redesign. Ex: Add new departments, handle increased admissions.

- NFR-6 Availability: 24/7 cloud access with minimal downtime and automatic backup.

3.3 Data Flow Diagram:



User inputs student and enrollment data -> System validates and stores records -> Approval process routes data for verification -> Instructor assignment automation executes -> Reports and dashboards consume processed data for monitoring.

3.4 Technology Stack:

- UI Layer: Salesforce Lightning Experience

- Logic Layer: Salesforce Flows (Record-Triggered, Screen Flows)

- Database Layer: Salesforce Custom Objects (Student*c, Course*c, Instructor*c, Enrollment*c)

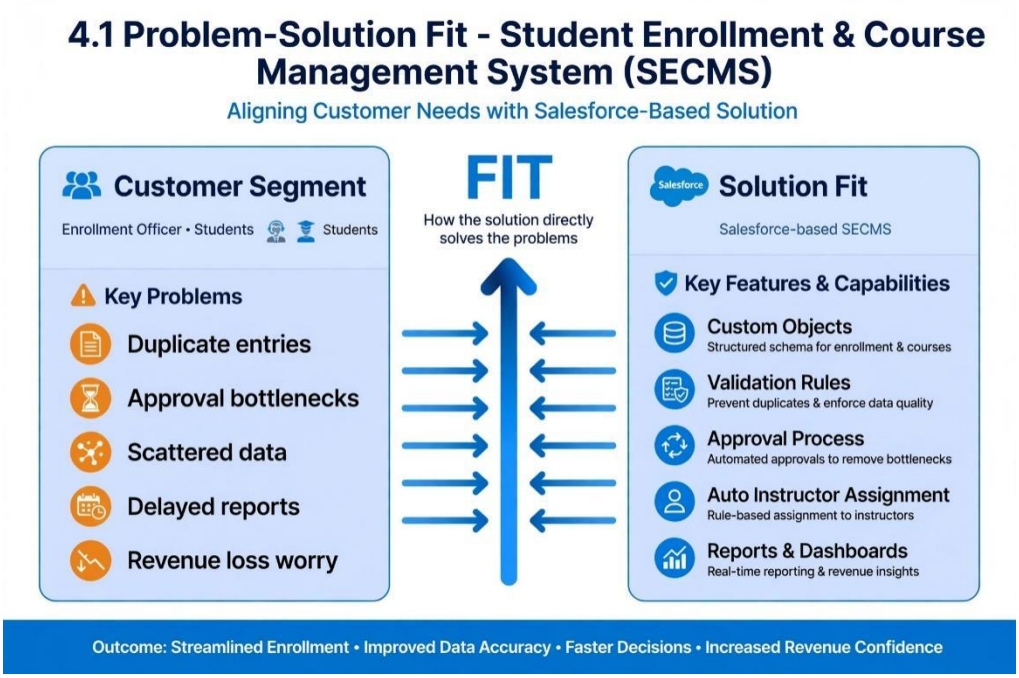
- Automation Layer: Validation Rules, Approval Process, Apex Triggers (if needed)

- Reporting Layer: Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit:

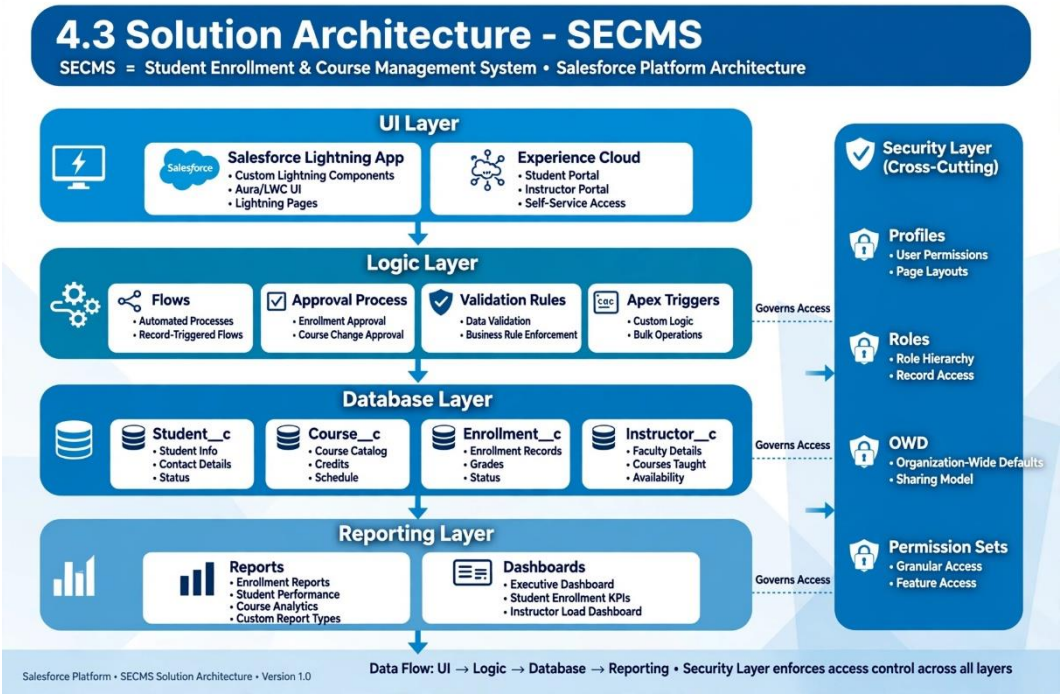
SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities. Manual pain points are mapped to Salesforce features to ensure feasibility and adoption.



4.2 Proposed Solution:

Parameter	Description
Problem	Manual, fragmented enrollment handling
Solution	Salesforce-based SECMS with end-to-end automation
Novelty	Automated instructor assignment, approval workflows, and task creation
Social Impact	Improved student experience and operational transparency
Revenue Model	Institutional deployment for universities/colleges
Scalability	Multi-department and multi-campus support

4.3 Solution Architecture:



The architecture integrates Salesforce Custom Objects linked via Lookup Relationships, automation via Flows and Approval Processes, security via Profiles/Roles/OWD, and analytics via Reports and Dashboards. All layers are native to Salesforce Cloud for high availability.

5. PROJECT PLANNING & SCHEDULING (Agile Methodology)

- Methodology: Agile - Epics -> User Stories -> Story Points -> Sprints

- Velocity calculation used for estimation

Here is *Activity 1: Creating Developer Account* — clean image + text for your documentation pa:

Step-by-Step to write in your DOC:

Product Backlog (Summary):

Sprint Epic US No User Story Points Priority

Sprint-1 Developer Setup USN-1 As an admin, I want to create a Salesforce developer org to configure SECMS 1 High

Sprint-1 Data Modeling USN-2 As an admin, I want to create custom objects for Student, Course, Enrollment, Instructor 3 High

Sprint-2 Data Modeling USN-3 As a user, I want dedicated tabs for easy navigation 3 High

Sprint-2 Data Modeling USN-4 As an admin, I want to create relevant fields for complete records 5 High

Sprint-2 Data Modeling USN-5 As an enrollment officer, I want a centralized Lightning App 3 High

Sprint-2 Data Modeling USN-6 Define relationships between Students, Courses, and Enrollments 5 Medium

Sprint-2 Data Modeling USN-7 Customized page layouts for clarity 5 High

Sprint-3 Automation USN-8 Validate input data to prevent incorrect records 3 High

Sprint-3 Automation USN-9 Automated flows for enrollment processing and notifications 3 High

Sprint-3 Automation USN-10 Approval process for enrollment verification 5 High

Sprint-4 Security USN-11 Configure role-based security 5 High

Sprint-5 Reports USN-12 Reports on enrollments, courses, instructor assignments 4 High

Sprint-6 Dashboards USN-13 Visual dashboards for enrollment trends and KPIs 4 Medium

Project Tracker: Sprint-1 (20 points, 6 Days, 24 Oct - 29 Oct 2022) completed. Velocity and Burndown Chart maintained for Sprint tracking.

6. PROJECT DEVELOPMENT PHASE - Implementation Milestones

Milestone 1: Salesforce Developer Account Creation

Activity 1: Creating Developer Org in Salesforce

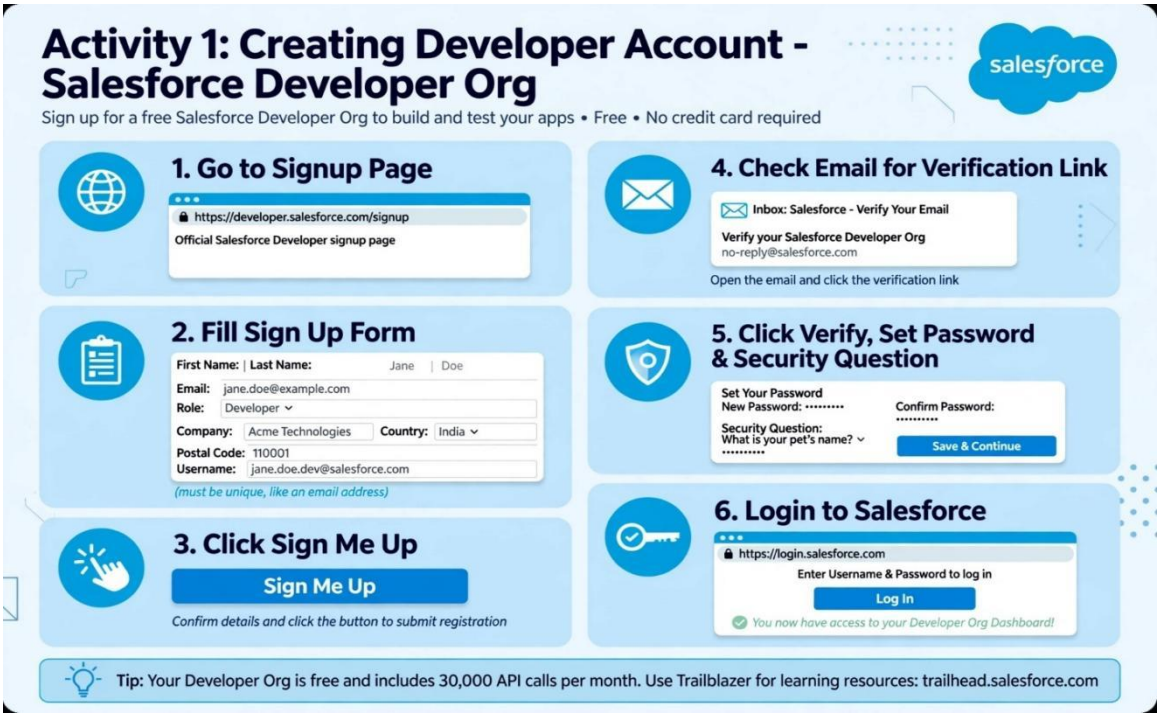
1. ***Go to signup link:* https://developer.salesforce.com/signup**

2. ***Enter the following details in the Sign-up form:***

    - **First Name: [Your First Name]**
    - **Last Name: [Your Last Name]**
    - **Email: [Your College Email]**
    - **Role: Developer**
    - **Company: [Your College Name]**
    - **Country: India**
    - **Postal Code: [Your Area Pincode]**
    - **Username: `yourname.dev@secms2026.com` (Must be unique, format like an email)**

3. **Click *Sign Me Up***

4. **Check your Email Inbox -> Open email from Salesforce -> Click *Verify Account***



5. **Set Password + Security Question -> Click *Save***

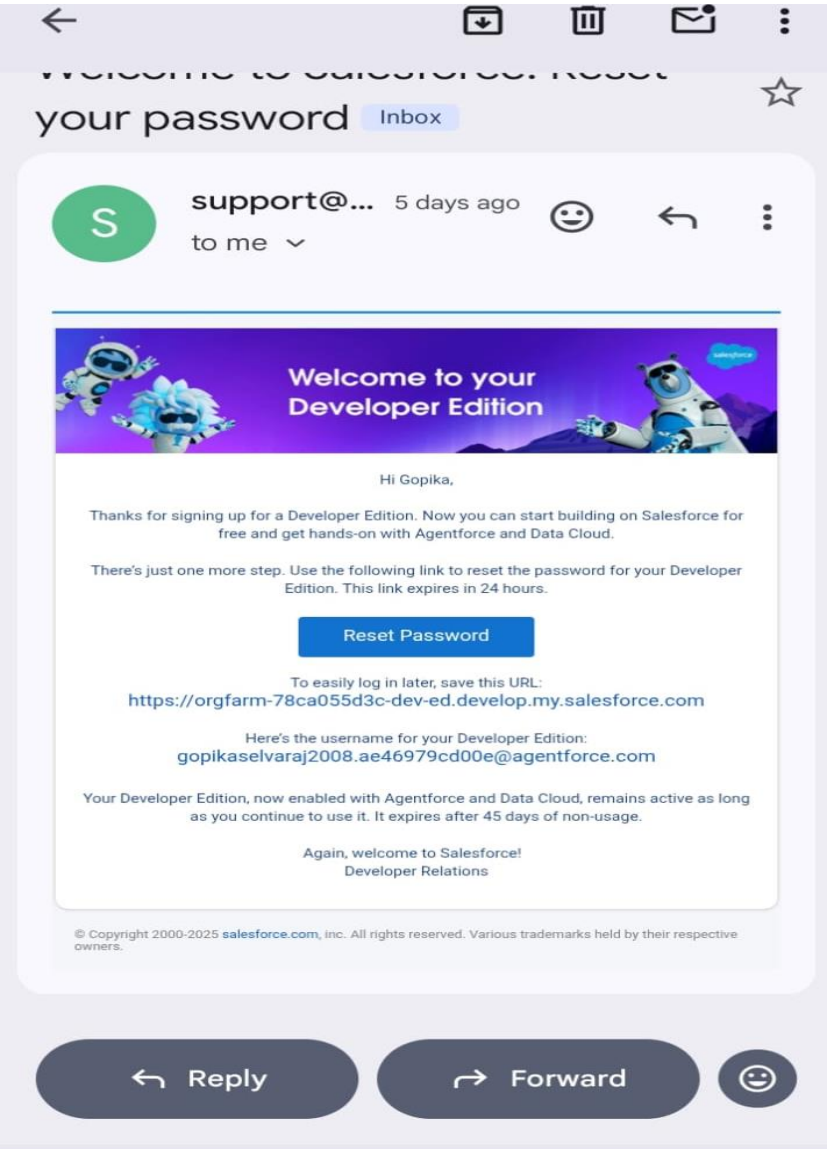
6. **Now login at https://login.salesforce.com with your Username & Password**

Activity: 2

Click on Verify AccountAfter signup, Salesforce will send a verification email to your registered email ID from no-reply@salesforce.comSubject: Verify your Salesforce

Developer OrgOpen the email and click on the Verify Account link — it will redirect to password set page.

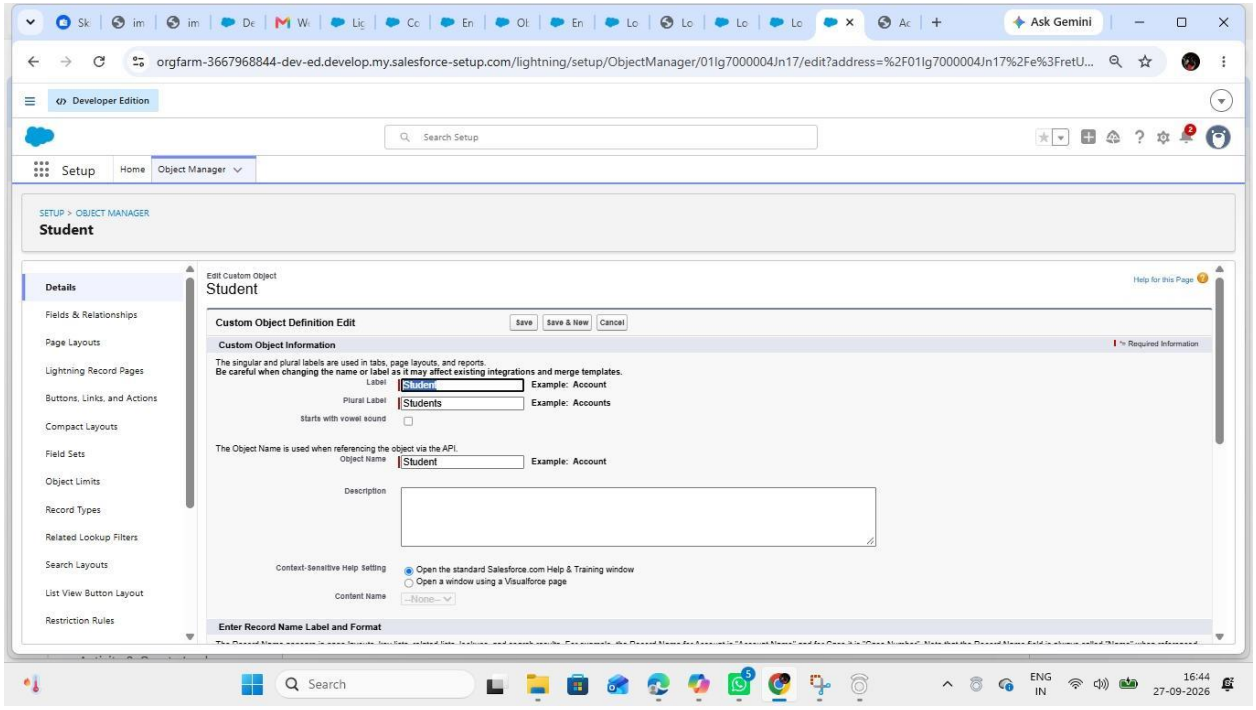
Create account at http://developer.salesforce.com, verify email, set password and security question to access Setup.

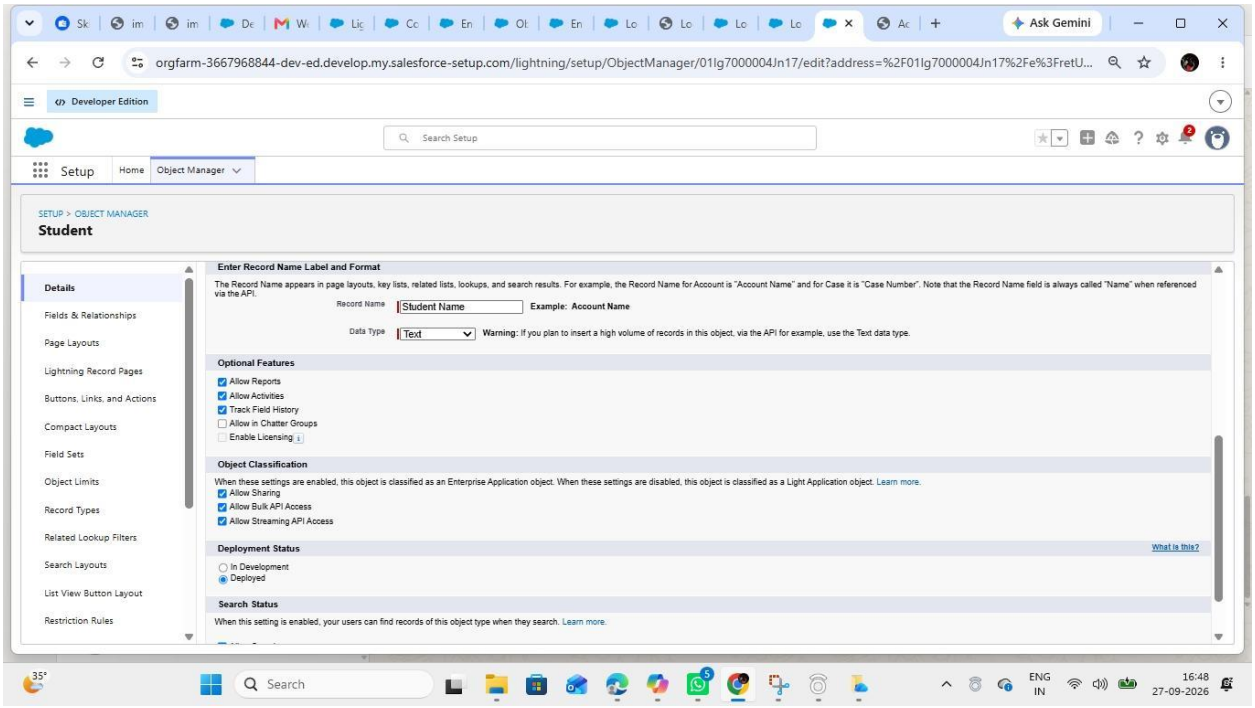


Milestone 2: Custom Object Creation

- Student__c (Student Name - Text), Allow Reports, Allow Search

- Course*c, Instructor*c, Enrollment__c - Created with same process via Object Manager -> Create -> Custom Object.





Milestone 3: Fields Creation

- Student*c: Email (Email), Phone (Phone), DOB (Date), Student_Status*c (Picklist: Active/Inactive/Alumni), Address (Text Area)

- Course*c: Course_Name*c (Text), Category*c (Picklist: Technical/Language/Non-Technical), Duration*c (Number), Course_Fee*c (Currency), Description*c (Text Area)

- Instructor*c: Instructor_Code*c (Text), Expertise__c (Picklist), Phone, Email

- Enrollment c: Student c (Lookup - Student), Course*c (Lookup - Course), Instructor c(Lookup - Instructor), Enrollment_Date*c (Date), Enrollment_Status c (Picklist: Requested/Approved/Rejected), Fees_Paidc (Checkbox), Total_Amount c (Currency), Comments__c (Text Area)

Table 1: Student Object Fields:

Field Label	Data Type
Student Name	Text (Standard)
Email	Email
Phone	Phone
Date of Birth	Date
Student Status	Picklist
Address	Text Area

Table 2: Course Object Fields:

Field Label	Data Type
Course Name	Text (Standard)
Category	Picklist

Table 3: Instructor Object Fields:

Field Label	Data Type
Instructor Name	Text (Standard)
Instructor Code	Text
Expertise	Picklist
Phone	Phone
Email	Email

Table 4: Enrollment Object Fields:

Field Label	Data Type
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date
Enrollment Status	Picklist
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Milestone 4: Relationships

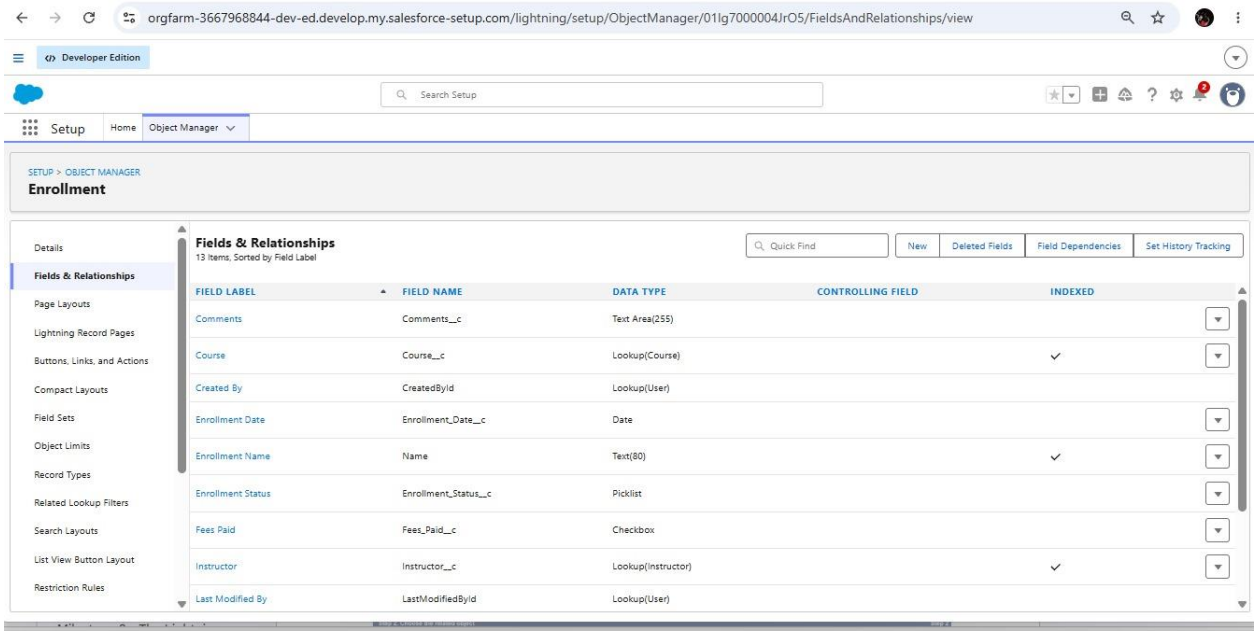
Lookup Relationships from Enrollment*c to Student*c, Course*c, and Instructor*c created via Object Manager -> Fields & Relationships -> New -> Lookup.

Create lookup for each object

1.lookup-> student

2.lookup->course

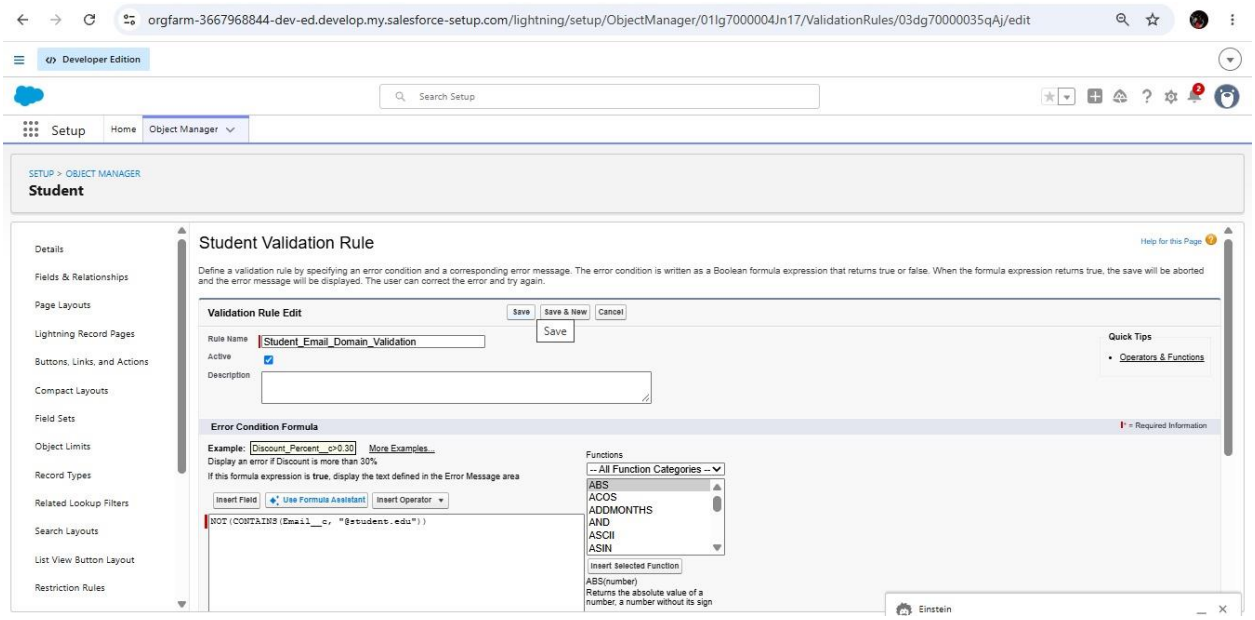
3.lookup->instructor

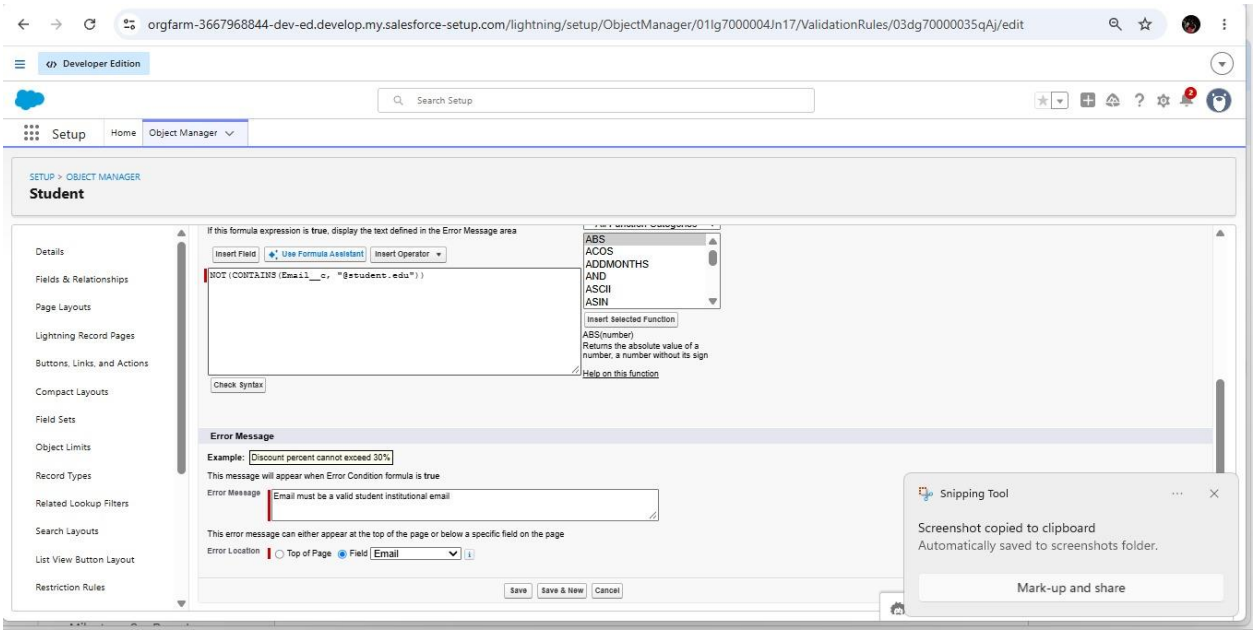


Milestone 5: Validation Rules

- Student Email Domain:`NOT(CONTAINS(Email__c, "@student.edu"))` - Error: Email must be valid institutional email.

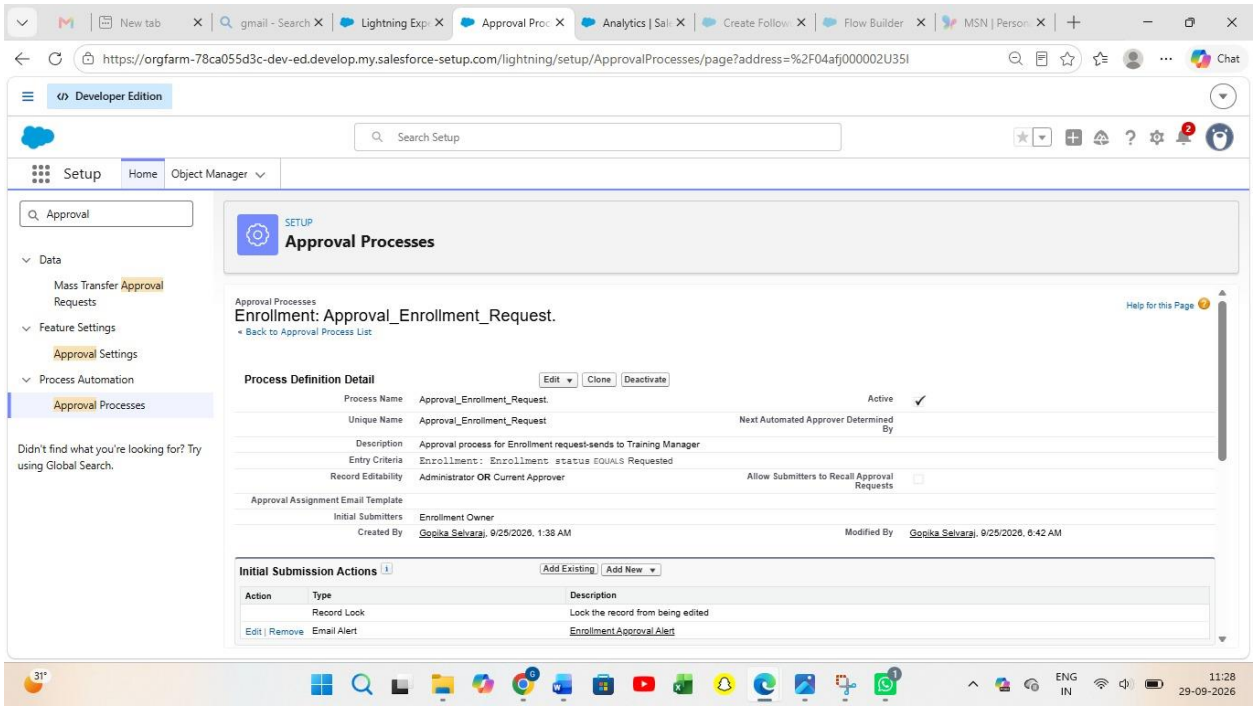
- Fees Must Be Paid Before Approval: `AND(ISPICKVAL(Enrollment_Status*c, "Approved"), Fees_Paid*c = FALSE)` - Error: Fees must be paid before approving.





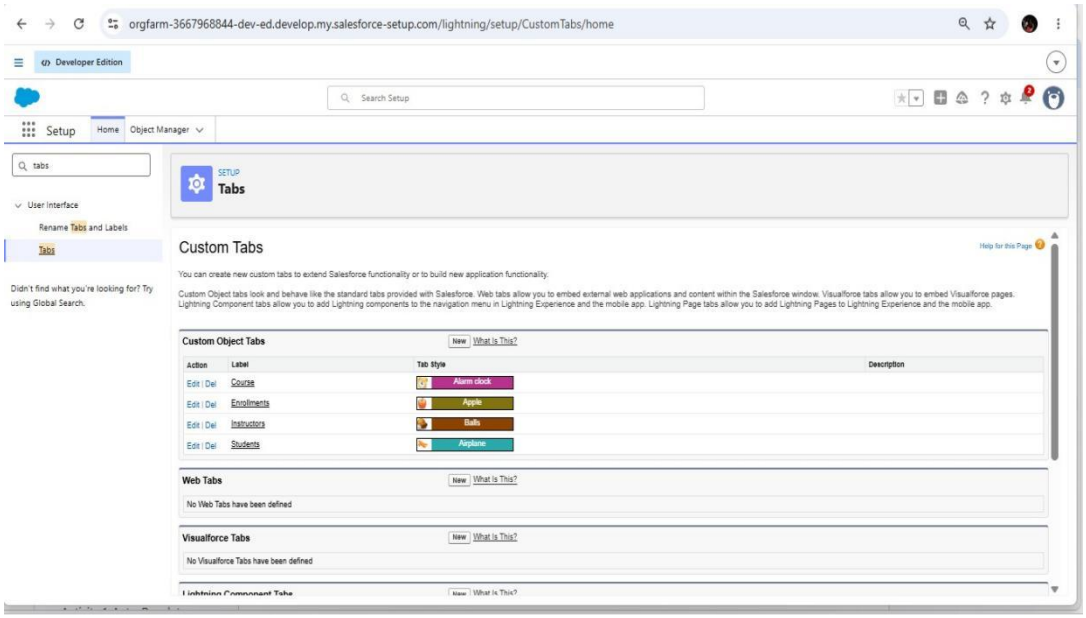
Milestone 6: Approval Process

Process Name: Approval_Enrollment_Request on Enrollment*c. Entry Criteria: Enrollment_Status*c = "Requested". Approver: Training Manager. Final Approval Action: Field Update to Approved. Final Rejection Action: Field Update to Rejected + Email Alert to student. Process Activated.



Milestone 7: Tabs Creation

Setup -> Tabs -> Custom Object Tabs -> New -> Select Student/Course/Instructor/Enrollment -> Assign tab style and profiles -> Save.



Create all tabs:

Course

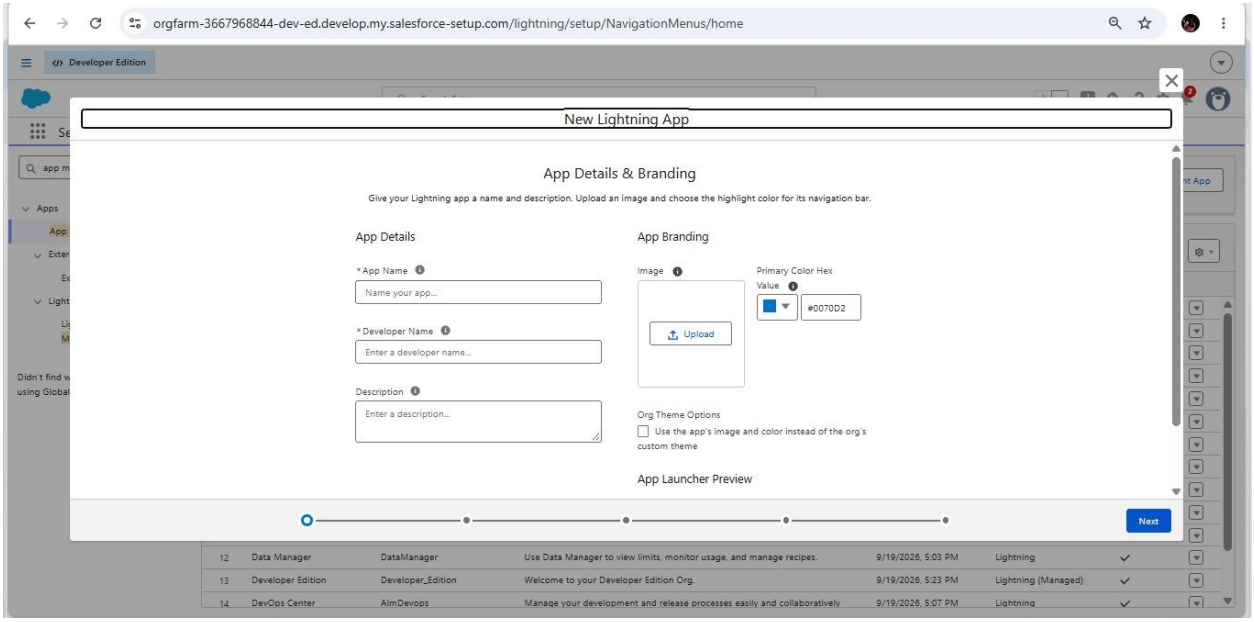
Enrollment

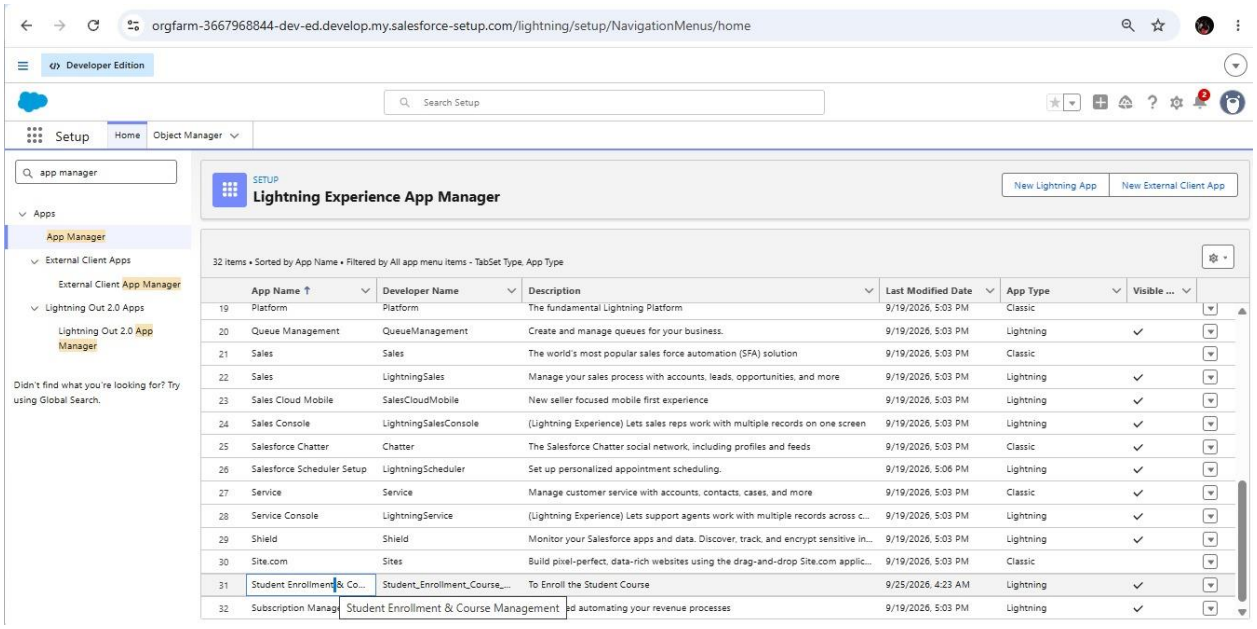
Instructor

Students

Milestone 8: Lightning App Creation

App Manager -> New Lightning App -> App Name: Student Enrollment & Course Management -> Standard Navigation -> Assign to Admin, Enrollment Officer, Instructor profiles -> Add tabs: Student, Course, Instructor, Enrollment, Reports, Dashboards, Home, Tasks -> Save & Finish.



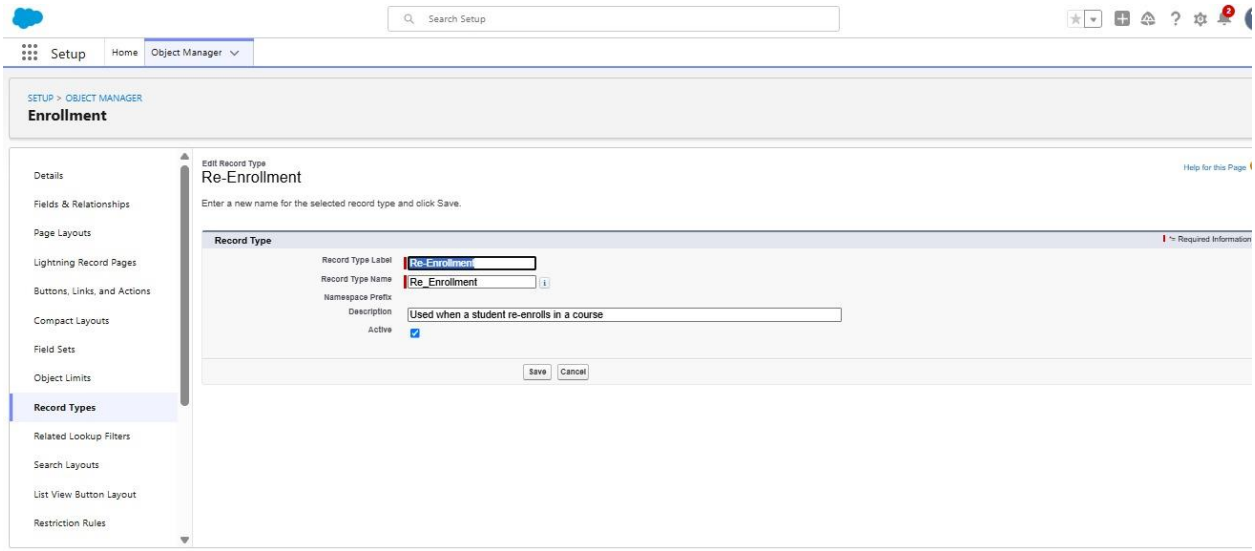


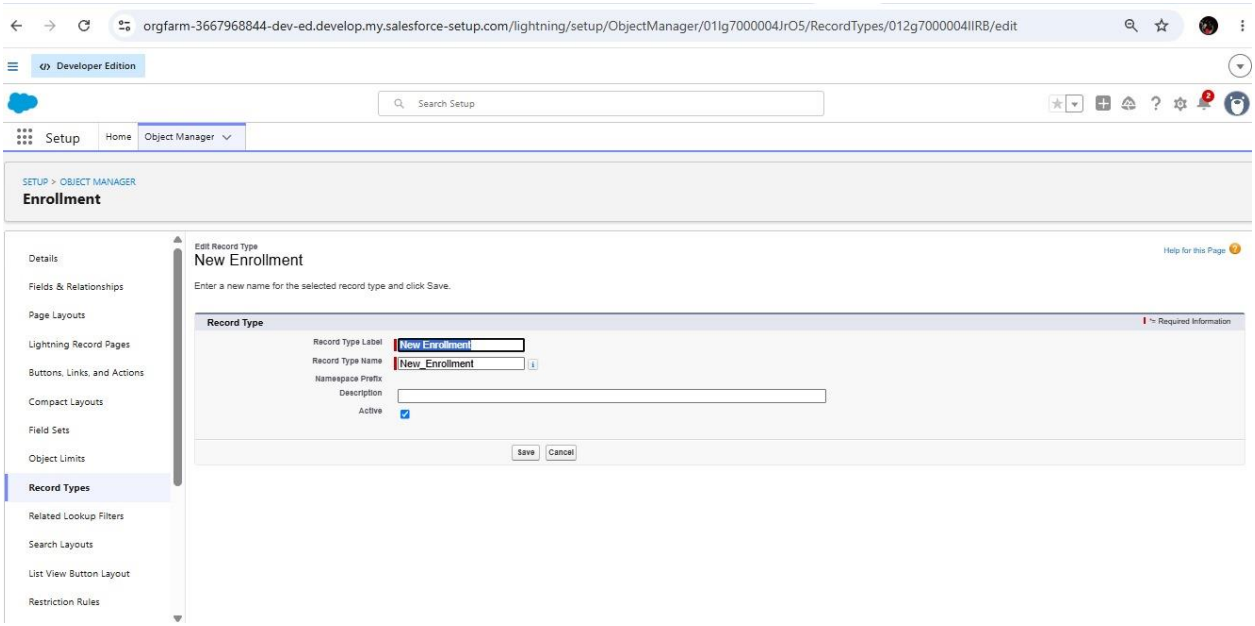
Milestone 9: Page Layouts and Record Types

- Record Types for Enrollment__c: New Enrollment and Re-Enrollment enabled for Admin and Enrollment Officer.

- Page Layouts: New Enrollment Layout and Re-Enrollment Layout cloned and customized with relevant fields.

- Related Lists: Add Enrollments related list to Student, Course, and Instructor page layouts.





Create :

Re enrollment

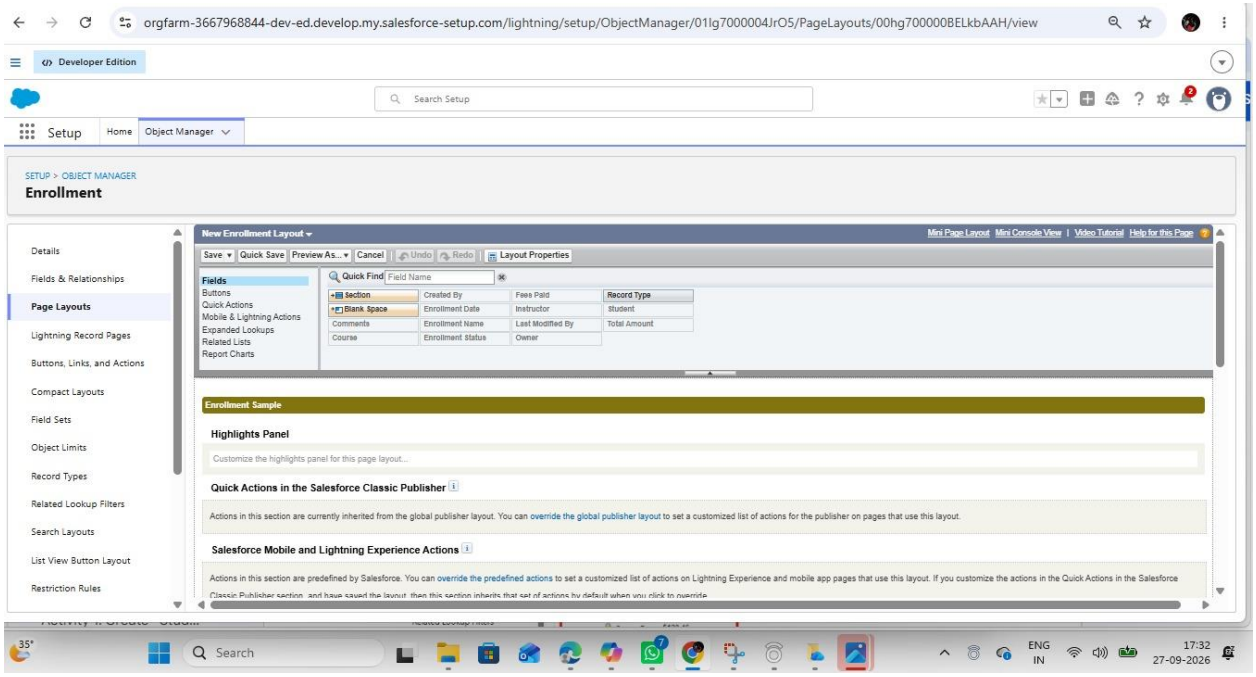
New enrollment

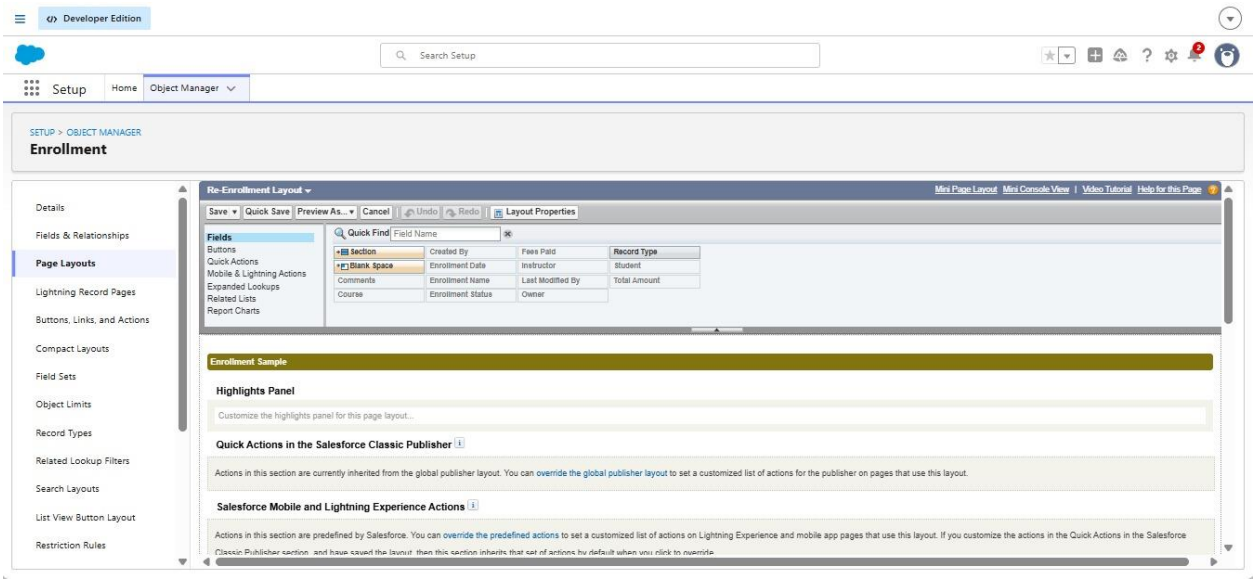
Step 2:

Create layout for

Re enrollment

New enrollment





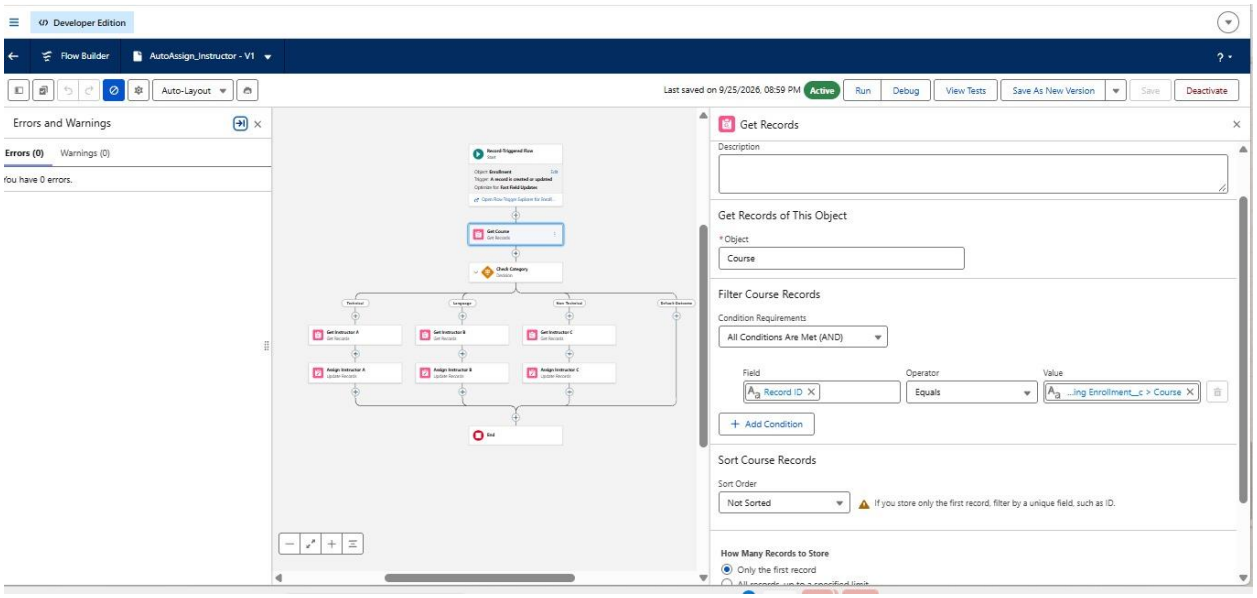
Milestone 10: Automation with Flows

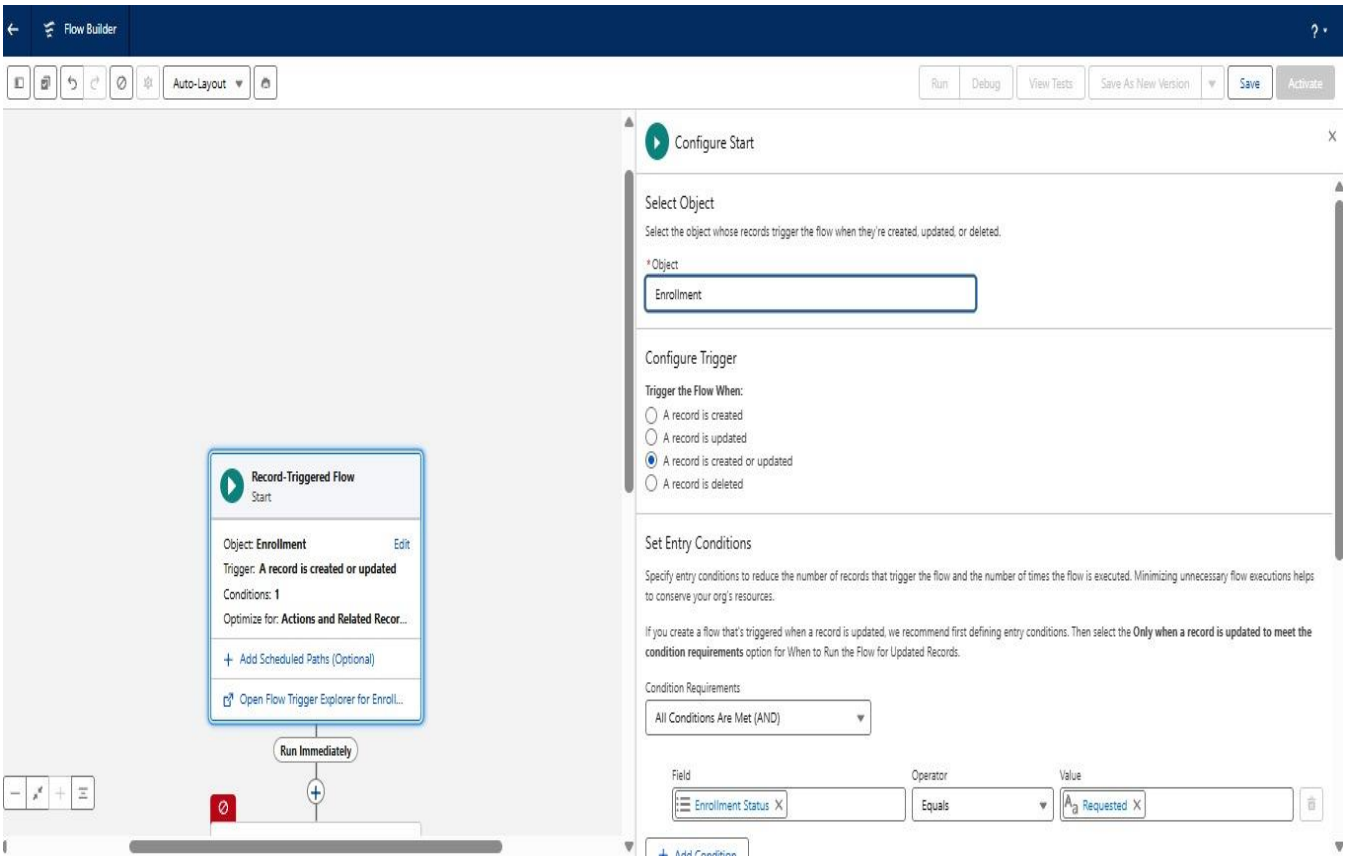
- Flow 1 - Auto-Populate Enrollment Date:* Record-Triggered Flow on Enrollment*c (on Create) -> Update Enrollment_Date*c = $Flow.CurrentDate.

- Flow 2 - Email on Approval:* Record-Triggered Flow on Update when Status = Approved -> Get Student -> Email Alert to student, Update Student Status to Active.

- Flow 3 - Auto-Assign Instructor:* Record-Triggered Flow on Enrollment__c -> Get Course -> Decision on Course Category (Technical -> Instructor A, Language -> Instructor B, Non-Technical -> Instructor C) -> Get Instructor record -> Update Enrollment Instructor.

- Flow 4 - Auto-Create Task:* Record-Triggered Flow when Status = Requested -> Get Admin User -> Create Task: Subject = Follow up on enrollment request, Due Date = TODAY()+2, Owner = Admin, WhatId = Enrollment Id.





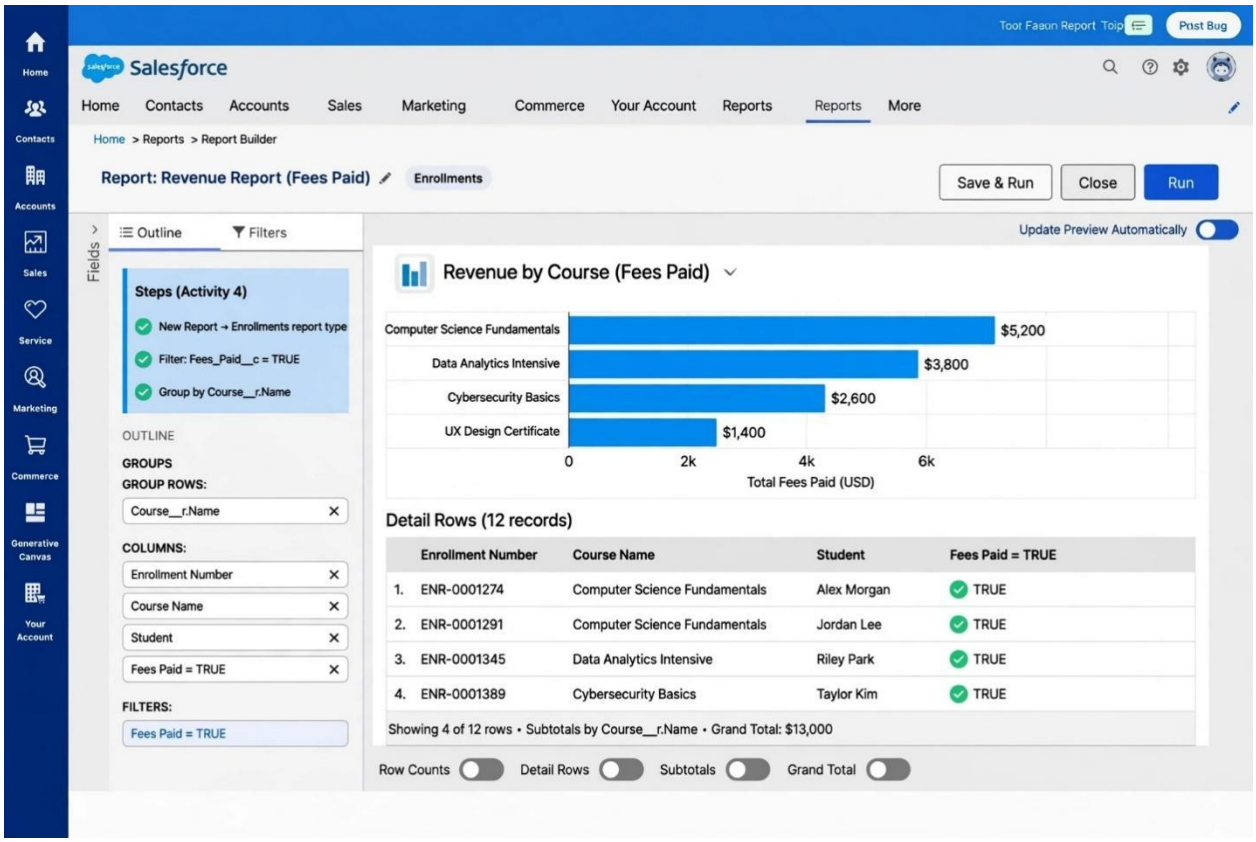
Milestone 11: Reports

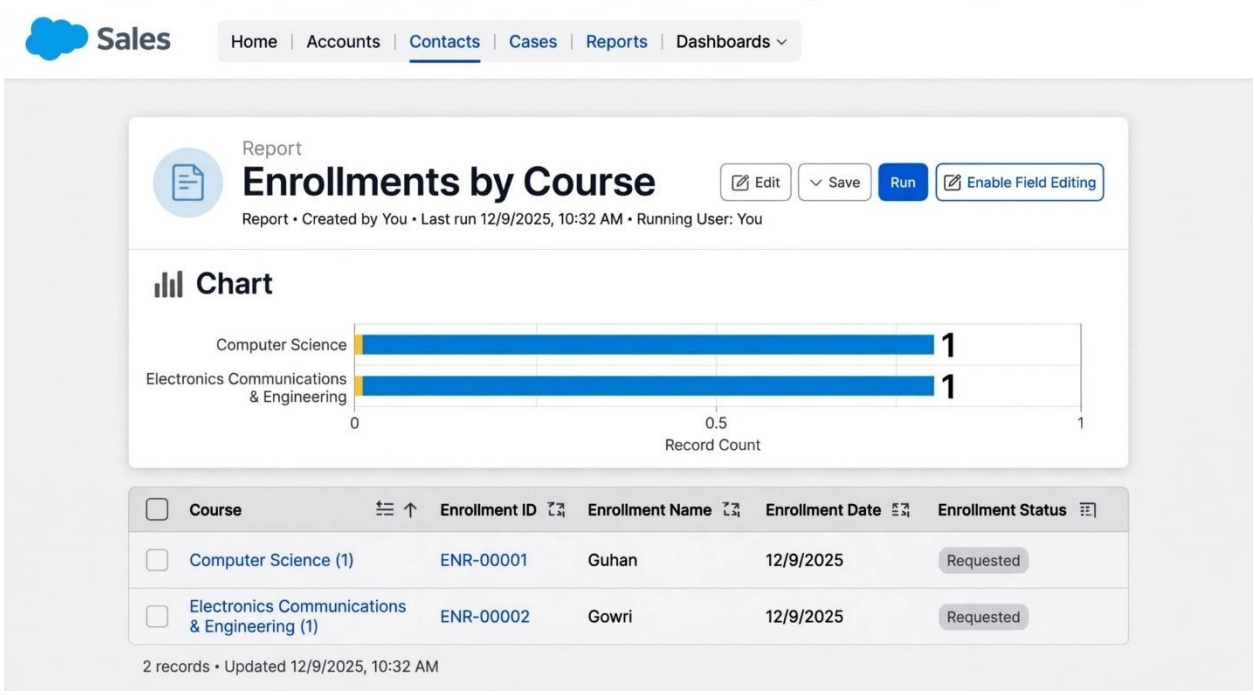
- Students by Status: Group by Status__c

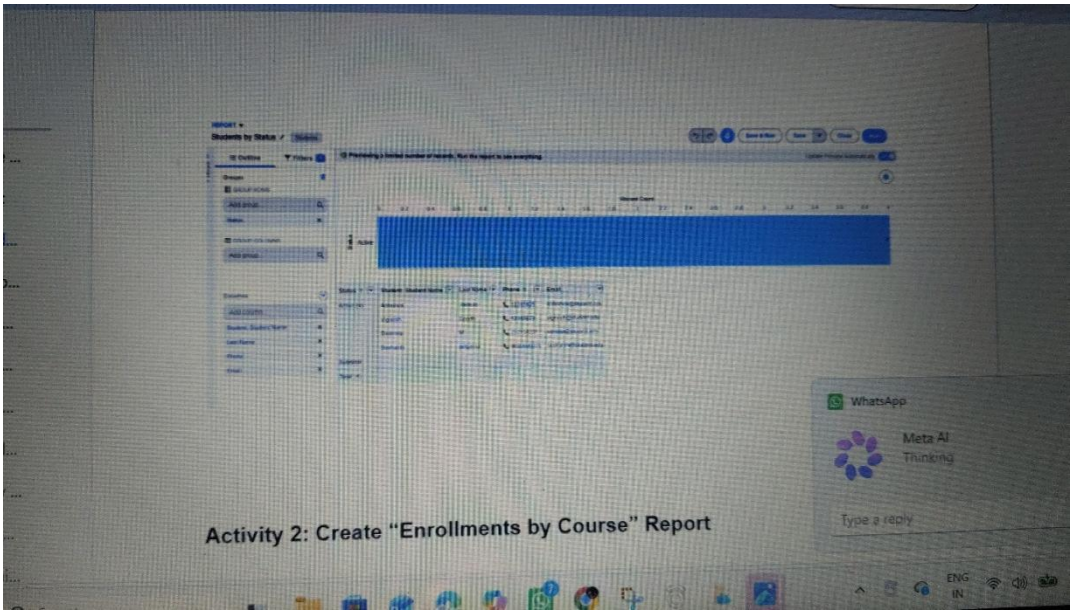
- Enrollments by Course: Group by Course__r.Name, Show Count

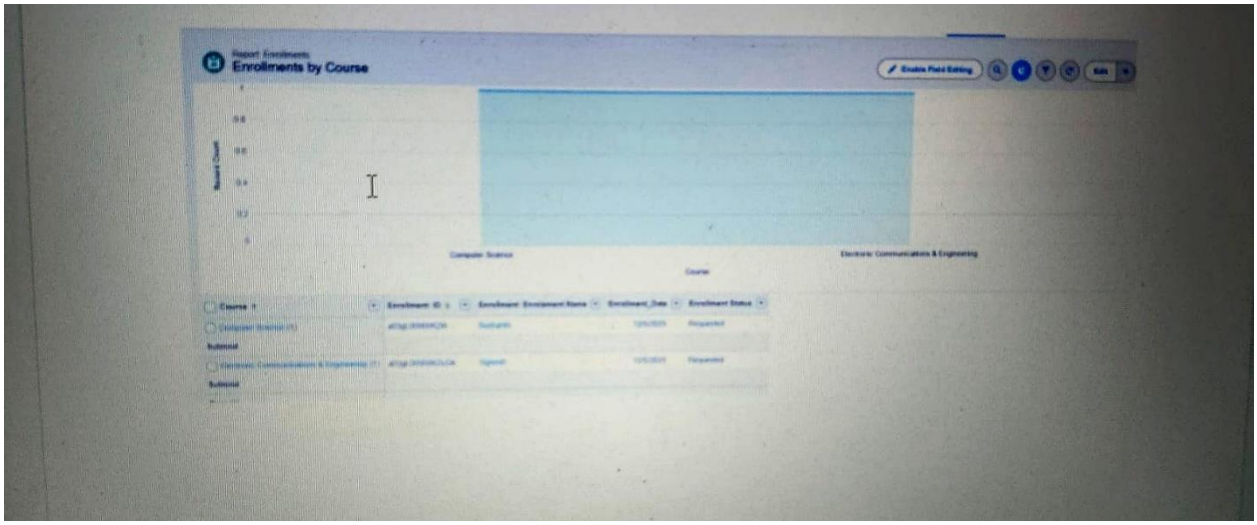
- Pending Enrollment Approvals: Filter Status = "Requested"

- Revenue Report (Fees Paid): Filter Fees_Paid*c = TRUE, Group by Course, SUM Total_Amount*c



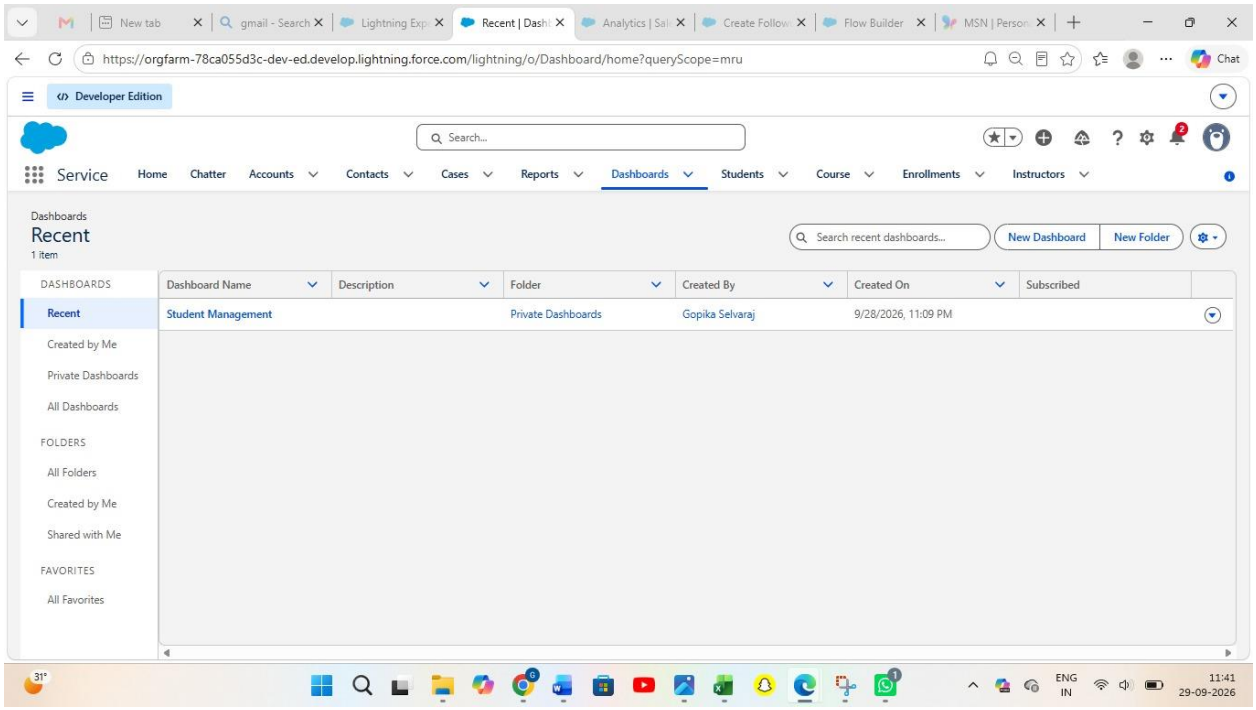

Sales
Home | Accounts | Contacts | Cases | Reports | Dashboards
Report
Enrollments by Course
Edit
Save
Run
Enable Field Editing
Report • Created by You • Last run 12/9/2025, 10:32 AM • Running User: You
Chart
Computer Science
1
Electronics Communications & Engineering
1
0
0.5
1
Record Count
Course
Enrollment ID
Enrollment Name
Enrollment Date
Enrollment Status
Computer Science (1)
ENR-00001
Guhan
12/9/2025
Requested
Electronics Communications & Engineering (1)
ENR-00002
Gowri
12/9/2025
Requested
2 records • Updated 12/9/2025, 10:32 AM





Milestone 12: Dashboards

Dashboard Name: Student Management Dashboard. Components: Donut Chart (Students by Status), Bar Chart (Enrollments by Course), Gauge (Total Revenue), Table (Pending Approvals).



Milestone 13: Security Setup

- *Profiles:* Admin (Full Access), Enrollment Officer (Read/Create/Edit on Enrollment, Read on Student/Course), Instructor (Read-only on assigned Enrollments).

- *Permission Sets:* Custom Object Access, Dashboard Editor.

- *OWD:* Student*c - Public Read/Write, Course*c - Public Read/Write, Enrollment__c - Private. For instructor access, Flow sets OwnerId = Instructor User.

7. FUNCTIONAL & PERFORMANCE TESTING

Screenshots captured to verify record creation, validation rule enforcement, record type behavior, flow execution for approval and seat checks, Apex trigger confirmation, and reports/dashboards accuracy. Testing confirms data accuracy, workflow automation, and performance monitoring.

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Centralized student, course, and enrollment data

- Automated workflows reducing manual effort and errors

- Real-time reporting and performance monitoring

- Role-based security ensuring data protection

- Scalable and customizable for future growth

Disadvantages:

- Salesforce platform dependency and licensing cost

- Requires initial configuration effort and user training

9. CONCLUSION

The Salesforce Student Enrollment & Course Management System successfully streamlines the entire student academic lifecycle from registration to completion. By leveraging Salesforce automation tools such as Validation Rules, Flows, Approval Processes, and Dashboards, the system improves data accuracy, operational efficiency, and decision-making. This project demonstrates how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

10. FUTURE SCOPE

- Integration with payment gateways (Razorpay, Stripe)
- Student self-service portal via Experience Cloud
- AI-driven analytics and enrollment predictions
- Mobile access support via Salesforce Mobile App

11. APPENDIX

- Source Code: Salesforce Configuration (Flows, Validation Rules, Approval Process)
- Dataset: Salesforce Custom Objects
- Demo: Salesforce Developer Org

Thank You
